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Advanced Seminar Data Science in Finance

Modelling Market Regimes: A comparison of rule-based and Gaussian Hidden Markov Model classification

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List of Abbreviations

HMM Hidden Markov Model. 2, III

S&P 500 Standard & Poor's 500 stock market index. 1, III

1 Introduction

The literature on Market Regimes is large and widely discussed in academia. What's especially interesting, is the connection between the Market Regime literature and findings in empirical finance on international market correlations. For example, it is a described phenomenon, that stock market volatility is higher when markets go down. Additionally, it is also well known that in periods of high volatility, stock markets become more correlated and in periods of low volatility, they are less correlated. As an exercise in this paper, the motivation is to implement a Hidden Markov Model (HMM) to infer market regimes from real financial data, by using the Python programming language. The research question this works intends to answer, is whether an HMM captures periods of elevated international equity market correlation.

2 Literature Review

2.1 Market Regimes in Financial Markets

Financial markets exhibit periods of stress, increased volatility and negative returns Oertmann (2026). Investors try to mitigate the risks arising from these movements, by diversifying their investments internationally Solnik (1974). However, in times of high volatility markets, empirical evidence suggests that correlations between international markets increase Ang and Bekaert (2002). This is especially relevant to international investors, because the benefits of diversification are lost.

The market conditions underlying such periods are difficult to measure because they are not directly observable. Instead, they must be inferred from the often times very noisy financial data. Hamilton (1989) introduced a Markov-switching framework for modeling unobserved changes in the dynamics of macroeconomic business-cycles. The same framework has subsequently been adapted to financial markets ?, where latent states are interpreted as market regimes characterized by different conditional return distributions, volatilities, and covariance structures.

Within the relevant literature, it is proposed to account for market regimes, in the context of portfolio construction Ang and Bekaert (2002). Typical indications for market regimes that can be classified as "bear" markets for example, include often periods of increased volatility, increased international market correlations or a decrease in expected returns.

2.2 Statistical Properties of Market Regimes

The empirical evidence suggests that financial markets exhibit nonstationarity, meaning that return distributions are not constant and change over time. In an analysis on the dynamics of the Standard & Poor's 500 stock market index (S&P 500), Stărică and Granger (2005) report that returns are primarily concentrated in shifts of the unconditional variance, suggesting that nonstationarities significantly affect the behavior of stock returns. Additionally, the findings also raise questions around persistence in stock returns, meaning whether past events influence current return dynamics, or if the observed nonstationarities are the primary drivers of persistence.

2.3 Methods for Regime Identification

As stated in the sections above, market regimes are difficult to define precisely, but the literature generally refers to periods of distinct statistical properties in financial markets. Therefore, the central idea behind identifying a market regime is to classify certain statistical properties to a specific period in time. The financial literature on how to model market regimes is plentiful, but for the purpose of this seminar paper two categories are presented for classification: a rule-based and a probabilistic approach.

2.3.1 Rule-based Regime Identification

A heuristic example for a rule-based approach to market regime classification is taken from a lecture (Oertmann, 2026): Stock market returns are subdivided into up- and downmarkets. A month is classified as upmarket if the respective return is above average and the reverse is true for downmarkets. Given a series of m up- and n downmarkets, the following up and down-volatility measures are computed with equations (1) and (2) respectively:

$$\sigma_{up} = \sqrt{\frac{1}{m-1} \sum_{t=1}^m (r_{up,t} - m r_{full})^2} \quad (1)$$

$$\sigma_{down} = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_{down,t} - n r_{full})^2} \quad (2)$$

2.3.2 Gaussian Mixture Models

While the above example is sufficient to gain an intuitive grasp on the classification of market regimes, a more sophisticated approach is provided by

mixture models. Mixture models are probabilistic models that introduce latent (or hidden) variables, allowing complex probability distributions over observed variables to be represented by a combination of simpler underlying distributions (Bishop, 2006a). This fits especially well, considering the fact that financial returns over long periods of time display fat tails and multimodality (**need a citation here**). A well-known mixture model is the Gaussian Mixture Model (GMM), which represents the observed data as a weighted combination of Gaussian probability distributions. Within the finance literature, GMMs have been applied to identifying "correlation breakdowns", "market contagion", and "volatility shocks" (Dixon et al., 2020a). The model assumes that observations are drawn independently from a mixture of K Gaussian components (or states) and is formally defined as (Bishop, 2006b):

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k) \quad (3)$$

Where $\pi = \{\pi_1, \dots, \pi_K\}$ represents a set of weights, given to each normally distributed component (or state) of K (Dixon et al., 2020a). Each Gaussian density $\mathcal{N}(x|\mu_k, \Sigma_k)$ is called a component of the mixture and has its own mean μ_k and covariance and Σ_k (Bishop, 2006b). In theory, each of these distributions could be interpreted to represent a market regime by defining a gaussian distribution for each state of the market.

2.3.3 Hidden Markov Models

A widely discussed approach in practical papers is the use of probabilistic sequence modeling Dixon et al. (2020b). In this school of thought, it is assumed that a latent (or unobservable) process evolving over time X_t , drives another, observable process Y_t , which can be observed either at all times or at some subset of times (Dixon et al., 2020b). Such a model is also referred to as a state-space model and it consists of describing the evolution of a latent state over time and the dependence of the observables on the latent state. The model assumes a sequence of states $\mathbf{s} := \{s_t\}_{t=1}^T$ and a sequence of observations $\mathbf{y} = \{y_t\}_{t=1}^T$. Then, the joint distribution is given as the product of transition probability densities $p(s_t|s_{t-1})$ and emission probability densities, $p(y_t|s_t)$. Formally, this joint distribution is defined as:

$$p(\mathbf{s}, \mathbf{y}) = p(s_1)p(y_1|s_1) \prod_{t=2}^T p(s_t|s_{t-1})p(y_t|s_t) \quad (4)$$

This approach is interesting because unlike the above rule-based example, or the GMM, it also models the empirical phenomenon of persistence, meaning that the hidden state of an observation is dependent on the hidden state of the preceding observation. The state-space model widely used in the literature is the HMM. The model parameters are estimated using the *Baum-Welch* Algorithm, an expectation-maximization (EM) algorithm for maximum likelihood estimation. After parameter estimation, the so called *Viterbi Algorithm* is applied, to infer the most likely sequence of hidden states (Dixon et al., 2020b).

Compared to the Gaussian Mixture Model, the principal distinction of the Hidden Markov Model lies in the treatment of the latent state. While a GMM assumes that observations are conditionally independent and assigns each observation to one of several Gaussian components independently of previous observations. A Hidden Markov Model, however, assumes that the latent state of the currently observable variable evolves according to the latent state in the preceding time period.

2.4 Portfolio construction under consideration of Market Regimes

Constructing a Portfolio under

2.4.1 Mean-Variance Portfolio

2.5 Summary and Research objective

The literature reviewed above establishes three points relevant to this paper. First, international diversification reduces portfolio risk when correlations between markets are low, but these benefits seem to weaken during periods of market stress because correlations tend to increase (Solnik, 1974; Ang and Bekaert, 2002). Second, Financial return processes exhibit nonstationarity which means that their statistical properties, particularly volatility and covariance structures, may change over time (Stărică and Granger, 2005). Third, market regimes are not directly observable and may therefore be represented as latent states inferred from observed data. In contrast to a rule-based classification of up- and downmarkets, a Hidden Markov Model allows these latent states to evolve persistently through a Markov transition process. A GMM could potentially function as a more efficient clustering method, compared to a simple rule-based approach.

Essentially, there are several ways to model market regimes. The basic idea behind such a modelling approach is to categorize certain statistical qualities of a market, to a specific period in time. The advantage of using Markov Models, is that the empirical phenomenon of persistence is also taken into account, compared to a rule-based approach or a GMM, that does not account for temporal dependence. Despite the similarities, Gaussian Mixture Models differ fundamentally from Hidden Markov Models. A GMM assumes that observations are conditionally independent, and therefore does not model any temporal dependence between successive observations. Instead, each observation is assumed to be drawn from one of the K underlying Gaussian distributions. Within the Machine Learning literature, to determine the number of states needed to model the data, it is suggested to either use a Kolmogorov-Smirnoff (KS) test or to base it on entropy criteria (Dixon et al., 2020a). In theory, each of these distributions could be interpreted to represent a market regime by defining a gaussian distribution for each state of the market. However, the important distinction is that these distributions are independent and therefore, explicitly omit the temporal dependency that is modeled with a HMM. Within the finance literature, GMMs have been applied to identifying adjacent topics like "correlation breakdowns", "market contagion", and "volatility shocks" (Dixon et al., 2020a). These applications are closely related to market-regime identification, although they do not explicitly model the temporal evolution of regimes.

These findings motivate the research question, whether modelling persistence improves the identification of financially meaningful market regimes.

3 Methodology

In this seminar paper, the following universe of national stock markets are selected for the analysis:

Country	Index	Ticker
France	CAC 40	FCHI
Germany	DAX 40	GDAXI
Italy	FTSE MIB	FTSEMIB.MI
Netherlands	AEX	AEX
Spain	IBEX 35	IBEX

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